

Proposal: Smart-City Traffic Analysis using NYC Taxi Trips

CS F415 Data Mining — Group 5

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1. Executive Summary

The Municipal Corporation needs data-driven interventions to reduce congestion and improve transit planning. We analyze the NYC Taxi Trip Duration dataset with **1,458,644** trips from **January 1, 2016 to June 30, 2016**. The dataset has 11 features, including pickup/dropoff timestamps, GPS coordinates, passenger count, and trip duration.

We formulate three linked mining tasks: (1) *trip duration prediction*, (2) *spatial demand clustering*, and (3) *congestion-level classification*. Together, these support operational planning, infrastructure prioritization, and policy design.

2. Data Exploration Summary

The data is complete (no missing values) and temporally consistent (no pickup-after-dropoff records; no mismatch between computed and provided durations). Profiling shows strong right-skew in trip duration and clear time-of-day effects in demand and speed.

Key statistics

Metric	Value
Total records	1,458,644
Features	11
Date range	2016-01-01 to 2016-06-30
Mean / median trip duration	959.49 s / 662 s
Trip duration standard deviation	5,237.43 s
Mean / median Haversine distance	3.441 km / 2.094 km
Median estimated speed (overall)	12.792 km/h
Median speed: peak vs off-peak	11.851 vs 19.180 km/h
Trips during peak hours (7–9, 17–19)	30.69%
Vendor split (1 / 2)	46.50% / 53.50%

Data quality findings

- Coordinates outside NYC bounds (40.4–41.0 N, -74.3 to -73.7 W): **985 trips** (0.0675%); invalid pickups: 247 and invalid dropoffs: 971.
- Duration outliers include **8,595** short trips (<60 s, 0.5892%) and **4** extreme long trips (>24 h).
- Passenger-count anomalies are limited: **60** zero-passenger trips and **5** trips with more than 6 passengers.
- Derived-metric checks show **175** speed anomalies (>100 km/h) and **5,897** zero-distance trips.

Notable patterns

- Demand peaks during morning and evening commute windows.
- Median speed declines during those peak windows.
- Spatial demand concentrates in core Manhattan and airport corridors.

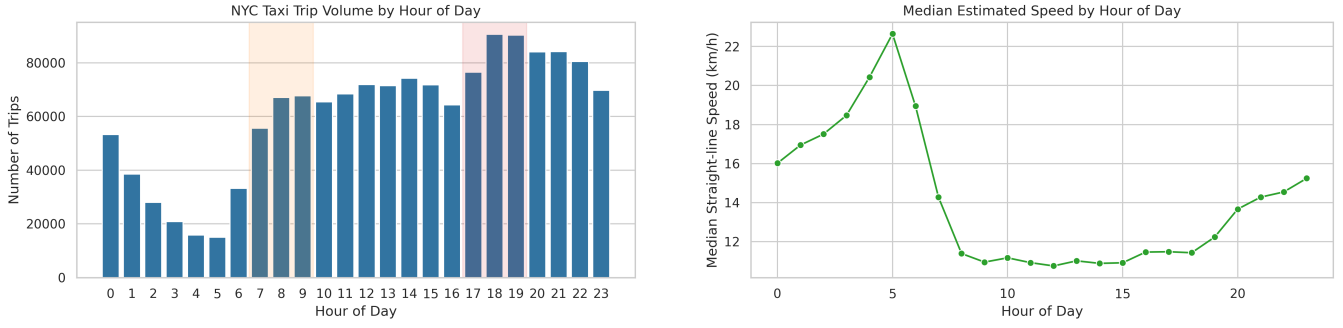


Figure 1: Hourly trip demand (left) and median straight-line speed (right) across the day.

3. Problem Formulation

Task 1: Trip Duration Prediction

Mathematical task: Supervised Regression

Input $\rightarrow$ Output: Temporal features (hour, weekday, month), pickup/dropoff coordinates, vendor/passenger fields, and Haversine distance $\rightarrow$ predicted trip duration (seconds).

Business value:

- Travel-time forecasting for route planning and passenger information systems.
- Incident flagging when observed duration significantly exceeds model prediction.
- Dynamic bus/subway capacity planning on parallel high-load corridors.

Planned models and evaluation: Linear Regression, Random Forest, Gradient Boosting; performance reported with RMSE/MAE and error breakdown by time-of-day.

Task 2: High-Demand Zone Identification

Mathematical task: Unsupervised Clustering

Input $\rightarrow$ Output: Pickup coordinates (optionally enriched with demand counts and mean duration) $\rightarrow$ cluster labels for geographic demand zones.

Business value:

- Identify underserved high-demand areas for transit augmentation.
- Prioritize local interventions (signal timing, lane policy, and curb management).
- Guide investment decisions such as bike-share, park-and-ride, and zone planning.

Planned models and evaluation: K-Means **vs.** DBSCAN (required algorithm comparison); quality assessed using Silhouette Score and cluster interpretability.

Task 3: Congestion Level Classification

Mathematical task: Supervised Multi-class Classification

Input $\rightarrow$ Output: Zone ID, hour, weekday, historical trip count, and average speed $\rightarrow$ congestion class {High, Medium, Low}. Labels are derived using speed-percentile thresholds across zone-time buckets.

Business value:

- Dynamic transit service-frequency control by zone and time bucket.
- Commuter-facing congestion alerts, route suggestions, and ETA guidance.
- Operations and staffing planning with evidence for policy interventions.

Planned models and evaluation: Logistic Regression and Random Forest Classifier; performance measured using Accuracy, macro-F1, and class-wise error analysis.